

Assigned: September 14; Due: September 28.

Note: Supporting files (including the CSV files needed for this assignment) can be found in the Homeworks folder on Canvas.

True or False? [25 pts]

Consider the standard form polyhedron $\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$. Assume that $\mathbf{A}$ has dimensions $m \times n$ and that its rows are linearly independent. Let $\mathbf{c} \in \mathbb{R}^n$. Consider the problem of minimizing $\mathbf{c}^\top \mathbf{x}$ over the polyhedron $\mathcal{P}$. For each of the following statements, state whether it is true (with a short proof) or whether it is false (with a counterexample).

- (a) If $n = m + 1$, then $\mathcal{P}$ has at most two basic feasible solutions. [4 pts]
- (b) At every optimal solution, no more than m variables can be positive. [4 pts]
- (c) If there is an optimal solution, there exists an optimal solution with at most m positive variables. [4 pts]
- (d) If $\mathbf{x}$ and $\mathbf{y}$ are two distinct basic feasible solutions of $\mathcal{P}$, then

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$$

is not a basic feasible solution for every $\lambda \in (0, 1)$. [4 pts]

- (e) Suppose that $\mathbf{x}$ is an optimal, nondegenerate basic feasible solution. If some nonbasic variable has zero reduced cost, then $\mathbf{x}$ is not the unique optimal solution. [4 pts]
- (f) Let $\mathbf{d} \in \mathbb{R}^n$. Consider the problem of minimizing $\max\{\mathbf{c}^\top \mathbf{x}, \mathbf{d}^\top \mathbf{x}\}$ over $\mathcal{P}$. If this problem has an optimal solution, it must have an optimal solution that is an extreme point of $\mathcal{P}$. [5 pts]

Geometry, basic feasible solutions, and simplex [25 pts]

Consider the following linear optimization problem, where $\lambda \in \mathbb{R}$ is a parameter:

$$\begin{aligned} \max_{x_1, x_2} \quad & x_1 + \lambda x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 5, \\ & 2x_1 + x_2 \leq 8, \\ & x_1 + 3x_2 \leq 9, \\ & x_1, x_2 \geq 0. \end{aligned}$$

- (a) Draw the feasible region and identify all of its extreme points. For each extreme point, identify which of the three inequality constraints are active.
- (b) Rewrite the problem in standard form using slack variables s_1, s_2, s_3 . For each extreme point, give at least one corresponding basis. Are any of the resulting basic feasible solutions degenerate? Explain your answer geometrically and algebraically.
- (c) Determine the optimal solution as a function of λ . Identify all values of λ for which the problem has multiple optimal solutions. For each λ , characterize the complete set of optimal solutions.
- (d) Set $\lambda = 1$. Starting from the basic feasible solution associated with $x_1 = x_2 = 0$, perform simplex iterations until an optimal solution is obtained. At every iteration, report the entering variable, leaving variable, step size, and new basic feasible solution.
- (e) In one of the ratio tests in part (d), more than one constraint becomes tight simultaneously. Explain what this means for the resulting basic feasible solution. Does this imply that the simplex method fails to make progress in objective value?

Simplex method with upper bound constraints [30 pts]

Consider a problem of the form

$$\begin{aligned} \min \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{0} \leq \mathbf{x} \leq \mathbf{u}, \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ has linearly independent rows and $\mathbf{u} \in \mathbb{R}^n$ with $u_j > 0, \forall j = 1, \dots, n$. In this problem, we extend the classic simplex method to solve this form.

- (a) Let $\mathbf{A}_{B(1)}, \dots, \mathbf{A}_{B(m)}$ be m linearly independent columns of $\mathbf{A}$ (the “basic” columns). We partition the set of all $i \neq B(1), \dots, B(m)$ into two disjoint subsets L and U . We set $x_j = 0$ for all $j \in L$, and $x_j = u_j$ for all $j \in U$. We then solve the equation $\mathbf{A}\mathbf{x} = \mathbf{b}$ for the basic variables $x_{B(1)}, \dots, x_{B(m)}$. Recall that a basic solution is defined such that the active constraints span the whole space. Show that the resulting vector $\mathbf{x}$ is a basic solution. Also, show that it is nondegenerate if and only if $x_j \neq 0$ and $x_j \neq u_j$ for every basic variable x_j .
- (b) Assume that the basic solution constructed in part (a) is feasible. Let $\bar{\mathbf{c}}$ denote the vector of reduced costs. Let x_j be some non-basic variable such that $x_j = 0$ and $\bar{c}_j < 0$. We increase x_j by θ and adjust the basic variables from $\mathbf{x}_B$ to $\mathbf{x}_B - \theta \mathbf{B}^{-1} \mathbf{A}_j$. What is the largest possible value of θ ? How are the new basic columns determined?
- (c) Assume again that the basic solution constructed in part (a) is feasible. Let $\bar{\mathbf{c}}$ denote the vector of reduced costs. Let x_j be some non-basic variable such that $x_j = u_j$ and $\bar{c}_j > 0$. We decrease x_j by θ and adjust the basic variables from $\mathbf{x}_B$ to $\mathbf{x}_B + \theta \mathbf{B}^{-1} \mathbf{A}_j$. What is the largest possible value of θ ? How are the new basic columns determined?
- (d) The above two steps define a simplex method for the problem. Assuming that every basic feasible solution is nondegenerate, show that the cost strictly decreases with each iteration and that the method thus constructed terminates.

Problem: production optimization [20 pts]

You optimize weekly production and inventory planning for a facility over $T = 52$ weeks. The facility manufactures $n = 20$ different product types from $m = 500$ materials that come in limited supply each week (materials have to be utilized on a weekly basis and cannot be carried over from one week to the next). Each product type generates a fixed per-unit profit that remains constant over time. You incur a holding cost based on the number of items held in inventory at the end of each week. The unit holding cost varies over time, but is identical across all product types. You are provided with a weekly demand forecast for each product type. You need not meet the full demand each week. You aim to maximize net profitability.

You have access to the following data files:

- **requirements.csv**: A matrix of size 500×20 that indicates the (continuous) amount of each of the 500 materials required to manufacture one unit of each of the 20 product types.
 - **availability.csv**: A matrix of size 500×52 that indicates the number of units of each of the 500 materials that can be used in the manufacturing process during each of the 52 weeks.
 - **demand.csv**: A matrix of size 20×52 that indicates the number of units demanded for each of the 20 product types during each of the 52 weeks.
 - **profit.csv**: A vector of size 20 indicating the per-unit profit of each product type (in \$).
 - **holding.csv**: A vector of size 52 indicating the per-unit holding cost in each week (in \$).
- a. Formulate a linear optimization model for the problem. Please specify your decision variables, your objective function and your constraints carefully.
 - b. Implement your model computationally. Report the sales profit and holding cost.
 - c. Represent graphically (i) the number of units manufactured for each product type and each week; (ii) the number of product types that get manufactured in some positive quantity each week; and (iii) the number of units stored in inventory for each product type and each week. Check out the **Plots** package and/or the **StatsPlots** package for creating your graphs. Interpret and explain your results.
 - d. In week 30, how many of the 500 materials were fully utilized? For how many of the 20 product types was demand fully satisfied? Comment briefly on the dynamics at play—notably, which constraints are binding and why?